Q1 - 24 June - Shift 1

Let $A\left(\frac{3}{\sqrt{a}}, \sqrt{a}\right) a > 0$, be a fixed point in the xy-plane. The image of A in y-axis be B and the image of B in x-axis be C. If D(3 cos θ, a sin θ) is a point in the fourth quadrant such that the maximum area of ΔACD is 12 square units, then a is equal to ________.

Space for your notes:

Q2 - 24 June - Shift 2

Let the area of the triangle with vertices A(1, α), B(α, 0) and C(0, α) be 4 sq. units. If the point (α, –α), (–α, α) and (α^2, β) are collinear, then β is equal to

(A) 64 (B) –8

(C) –64 (D) 512

Space for your notes:

Q3 - 26 June - Shift 1

Let R be the point (3, 7) and let P and Q be two points on the line x + y = 5 such that PQR is an equilateral triangle. Then the area of ΔPQR is :

(A) $\frac{25}{4\sqrt{3}}$ (B) $\frac{25\sqrt{3}}{2}$ (C) $\frac{25}{\sqrt{3}}$ (D) $\frac{25}{2\sqrt{3}}$

Space for your notes:

Q4 - 27 June - Shift 1

In an isosceles triangle ABC, the vertex A is (6, 1) and the equation of the base BC is 2x + y = 4. Let the point B lie on the line x + 3y = 7. If (α, β) is the centroid ΔABC, then $15(\alpha+\beta)$ is equal to :

(A) 39 (B) 41

(C) 51 (D) 63

Space for your notes:

Q5 - 27 June - Shift 1

A rectangle R with end points of the one of its dies as (1, 2) and (3, 6) is inscribed in a circle. If the equation of a diameter of the circle is 2x – y + 4 = 0, then the area of R is ________.

Space for your notes:

Q6 - 28 June - Shift 1

A ray of light passing through the point P(2, 3) reflects on the x-axis at point A and the reflected ray passes through the point Q(5, 4). Let R be the point that divides the line segment AQ internally into the ratio 2 : 1. Let the co-ordinates of the foot of the perpendicular M from R on the bisector of the angle PAQ be (α, β). Then, the value of $7\alpha+3\beta$ is equal to _______.

Space for your notes:

Q7 - 28 June - Shift 2

Let a triangle be bounded by the lines $L_1 : 2x + 5y = 10$; $L_2 : -4x + 3y = 12$ and the line L_3, which passes through the point P(2, 3), intersect L_2 at A and L_1 at B. If the point P divides the line-segment AB, internally in the ratio 1 : 3, then the area of the triangle is equal to

Space for your notes:

(A) $\frac{110}{13}$ (B) $\frac{132}{13}$

(C) $\frac{142}{13}$ (D) $\frac{151}{13}$

Q8 - 29 June - Shift 1

The distance between the two points A and A' which lie on y = 2 such that both the line segments AB and A' B (where B is the point (2, 3)) subtend angle $\frac{\pi}{4}$ at the origin, is equal to :

Space for your notes:

(A) 10 (B) $\frac{48}{5}$

(C) $\frac{52}{5}$ (D) 3

Q9 - 29 June - Shift 2

The distance of the origin from the centroid of the triangle whose two sides have the equations $x - 2y + 1 = 0$ and $2x - y - 1 = 0$ and whose orthocenter is $\left(\frac{7}{3}, \frac{7}{3}\right)$ is:

(A) $\sqrt{2}$
(B) 2
(C) $2\sqrt{2}$
(D) 4

Space for your notes:

Answer Key

Q1 (8)	**Q2** (C)	**Q3** (D)	**Q4** (C)
Q5 (16)	**Q6** (31)	**Q7** (B)	**Q8** (C)
Q9 (C)			

Q1 (8)

$$A=\left(\frac{3}{\sqrt{a}},\sqrt{a}\right)$$

$$B=\left(\frac{-3}{\sqrt{a}},\sqrt{a}\right)$$

$$C=\left(-\frac{3}{\sqrt{a}},-\sqrt{a}\right)$$

Area of ACD

$$\frac{1}{2}\begin{vmatrix}\frac{3}{\sqrt{a}} & \sqrt{a} \\ -\frac{3}{\sqrt{a}} & -\sqrt{a} \\ 3\cos\theta & a\sin\theta \\ \frac{3}{\sqrt{a}} & \sqrt{a}\end{vmatrix}$$

$$\frac{1}{2}6\sqrt{a}(\cos\theta-\sin\theta)$$

$$3\sqrt{a}(\cos\theta-\sin\theta)$$

max values of function is $3\sqrt{a}\sqrt{2}$

$$3\sqrt{a}\sqrt{2}=12$$

$$2a=16$$

$$a=8$$

Q2 (C)

$$\frac{1}{2}\begin{vmatrix} \alpha & 0 & 1 \\ 1 & \alpha & 1 \\ 0 & \alpha & 1 \end{vmatrix} = \pm 4$$

$\alpha = \pm 8$

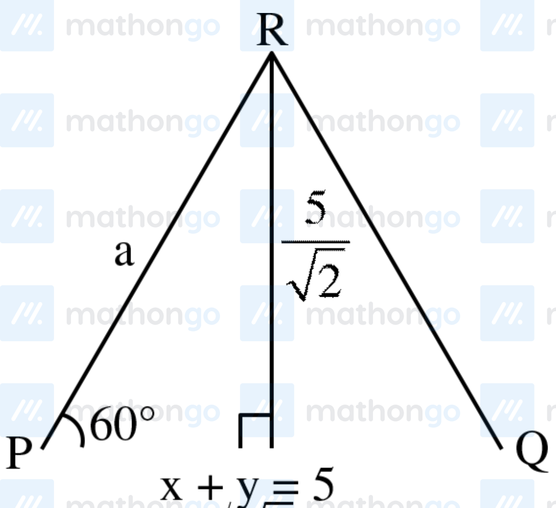
Now given points $(8, -8), (-8, 8), (64, \beta)$

OR $(-8, 8), (8, -8), (64, \beta)$

are collinear $\Rightarrow$ Slope $= -1$.

$\boxed{\beta = -64}$ **Ans. (C)**

Q3 (D)

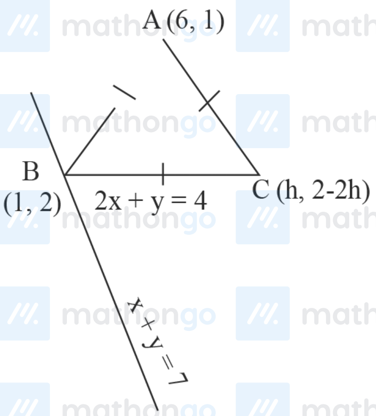


$$\sin 60° = \frac{5/\sqrt{2}}{a}$$

$$a = \frac{5\sqrt{2}}{3}$$

$$\text{Area of } \Delta PQR = \frac{\sqrt{3}}{4}a^2 \quad = \frac{25}{2\sqrt{3}}$$

Q4 (C)



Point B (1, 2)

Now let C be (h, 4 - 2h)

(As C lies on 2x + y = 4)

$\because \Delta$ is isosceles with base BC

$\therefore AB = AC$

$\sqrt{25+1} = \sqrt{(6-h)^2 + (2h-3)^2}$

$\sqrt{26} = \sqrt{36 + h^2 - 12h + 4h^2 + 9 - 12h}$

$26 = 5h^2 - 24h + 45 \Rightarrow 5h^2 - 24h + 19 = 0$

$\Rightarrow 5h^2 - 5h - 19h + 19 = 0$

$h = \frac{19}{5}$ or h = 1

Thus $C\left(\frac{19}{5}, \frac{-18}{5}\right)$

Centroid $\left(\frac{6+1+\frac{19}{5}}{3}, \frac{1+2-\frac{18}{5}}{3}\right)$

$\left(\frac{35+19}{15}, \frac{15-18}{15}\right)$

$\left(\frac{54}{15}, \frac{-3}{15}\right)$

$\alpha = \frac{54}{15}; \beta = \frac{-3}{15}$

$15(\alpha + \beta) = 51$

Q5 (16)

Eq. of line AB

$y = 2x$

Slope of AB = 2

Slope of given diameter = 2

So the diameter is parallel to AB

Distance between diameter and line AB

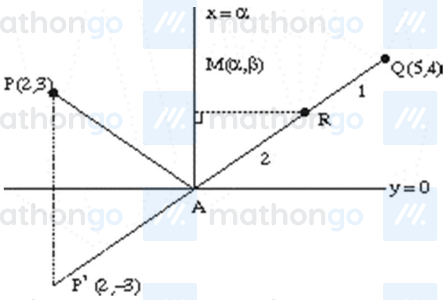
$$= \left(\frac{4}{\sqrt{2^2+12}}\right) = \frac{4}{\sqrt{5}}$$

Thus $BC = 2 \times \frac{4}{\sqrt{5}} = \frac{8}{\sqrt{5}}$

$$AB = \sqrt{(1-3)^2 + (2-6)^2} = \sqrt{20} = 2\sqrt{5}$$

Area = AB × BC = $\frac{8}{\sqrt{5}} \times 2\sqrt{5} = 16$ Ans.

Q6 (31)



By observation we see that $A(\alpha,0)$.

And β = y -cordinate of R

$$= \frac{2\times4+1\times0}{2+1} = \frac{8}{3} \quad \ldots(1)$$

Now P' is image of P in y = 0 which will be $P'(2,-3)$

$\therefore$ Equation of P'Q is $(y+3) = \frac{4+3}{5-2}(x-2)$

i.e. $3y+9 = 7x-14$

$A \equiv \left(\frac{23}{7},0\right)$ by solving with $y=0$

$\therefore\ \alpha = \frac{23}{7} \quad \ldots(2)$

By (1), (2)

$$7\alpha+3\beta = 23+8 = 31$$

Q7 (B)

Points A lies on L_2

$$A\left(\alpha, 4+\frac{4}{3}\alpha\right)$$

Points B lies on L_1

$$B\left(\beta, 2-\frac{2}{5}\beta\right)$$

Points P divides AB internally in the ratio 1 : 3

$$\Rightarrow P(2,3) = P\left(\frac{3\alpha+\beta}{4}, \frac{3\left(4+\frac{4}{3}\alpha\right)+1\left(2-\frac{2}{5}\beta\right)}{4}\right)$$

$$\Rightarrow \alpha = \frac{3}{13}, \beta = \frac{95}{13}$$

Point $A\left(\frac{3}{13}, \frac{56}{13}\right), B\left(\frac{95}{13}, -\frac{12}{13}\right)$

Vertex C of triangle is the point of intersection

L_1 & L_2

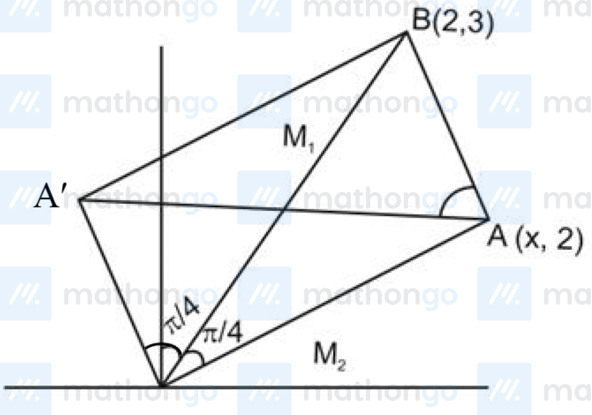
$$\Rightarrow C\left(-\frac{15}{13}, \frac{32}{13}\right)$$

$$\text{area } \Delta ABC = \frac{1}{2}\left\|\begin{matrix} \frac{3}{13} & \frac{56}{13} & 1 \\ \frac{95}{13} & -\frac{12}{13} & 1 \\ -\frac{15}{13} & \frac{32}{13} & 1 \end{matrix}\right\|$$

$$= \frac{1}{2\times 13^3}\left\|\begin{matrix} 3 & 56 & 13 \\ 95 & -12 & 13 \\ -15 & 32 & 13 \end{matrix}\right\|$$

$$\text{area } \Delta ABC = \frac{132}{13} \text{ sq. units.}$$

Q8 (C)



$M_1 = 3/2$ $M_2 = 2/x$

$$\tan\pi/4 = \left|\frac{3/2 - 2/x}{1 + 6/2x}\right| = 1$$

$\Rightarrow x_1 = 10, \quad x_2 = -2/5$

$\Rightarrow AA^1 = 52/5$

Q9 (C)

$AB \equiv x - 2y + 1 = 0$

$AC \equiv 2x - y - 1 = 0$

So A(1, 1)

Altitude from B is $BH = x + 2y - 7 = 0 \Rightarrow B\ (3, 2)$

Altitude from C is $CH = 2x + y - 7 = 0 \Rightarrow C\ (2, 3)$

Centroid of $\Delta ABC = E(2, 2)$ $OE = 2\sqrt{2}$